CHAPTER 8

RESPIRATION IN PLANTS

CHAPTER ESSENCE AT A GLANCE:
Sequential catabolic oxidation of respiratory substrates transfers chemical energy via electron carriers to generate ATP.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Glycolysis EMP Metabolic Pathway:** Glycolysis occurs in the cytoplasm of all living cells under anaerobic conditions. One molecule of glucose undergoes ten enzymatic steps to produce two molecules of pyruvic acid. The preparatory phase consumes 2 ATP (catalyzed by hexokinase and phosphofructokinase), while the pay-off phase generates 4 ATP and 2 NADH + 2H+, resulting in a net yield of 2 ATP and 2 NADH.

• **Fermentative Pathways and Anaerobic Yield:** Under anaerobic conditions, pyruvate undergoes fermentation to regenerate NAD+ for glycolysis: alcoholic fermentation in yeast utilizes pyruvate decarboxylase and alcohol dehydrogenase to yield ethanol and CO2; lactic acid fermentation in vigorous muscles uses lactate dehydrogenase to yield lactic acid without releasing CO2. Both extract less than 7% of substrate energy.

• **Link Reaction and the Krebs TCA Cycle:** In the presence of oxygen, pyruvate enters the mitochondrial matrix. The link reaction (oxidative decarboxylation by pyruvate dehydrogenase complex) converts pyruvate into Acetyl-CoA, NADH, and CO2. Acetyl-CoA condenses with oxaloacetate (OAA) to enter the Krebs cycle, generating 3 NADH, 1 FADH2, and 1 GTP (ATP) per acetyl group (doubled per glucose).

• **Mitochondrial ETS and Oxidative Phosphorylation:** The inner mitochondrial membrane houses five enzyme complexes of the Electron Transport System: Complex I (NADH dehydrogenase), Complex II (succinate dehydrogenase), Complex III (cytochrome bc1), Complex IV (cytochrome c oxidase with copper centers), and Complex V (ATP synthase). Oxygen acts as the final terminal electron acceptor forming water. Oxidation of 1 NADH yields 3 ATP, while 1 FADH2 yields 2 ATP.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$$\text{Glucose} + 2\text{NAD}^+ + 2\text{ADP} + 2\text{P}_i \longrightarrow 2\text{Pyruvate} + 2\text{NADH} + 2\text{H}^+ + 2\text{ATP}$$

Net Biochemical Stoichiometry of Glycolytic EMP Pathway

$$\text{RQ} = \frac{\text{Volume of } CO_2 \text{ evolved}}{\text{Volume of } O_2 \text{ consumed}} \quad [\text{Carbs} = 1.0,\ \text{Fats} = 0.7,\ \text{Proteins} = 0.9]$$

Respiratory Quotient Formulation and Standard Substrate Values

$$2C_{51}H_{98}O_6 + 145O_2 \longrightarrow 102CO_2 + 98H_2O \rightarrow \text{RQ} = \frac{102}{145} \approx 0.7$$

Respiratory Quotient Derivation for Tripalmitin Fat

$$\text{Net Aerobic ATP Yield} = \text{Glycolysis } (2) + \text{TCA } (2) + \text{ETS Oxidative Phosphorylation } (34) = 38 \text{ ATP}$$

Complete Aerobic Oxidation Energy Balance per Glucose Molecule

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The RQ value for carbohydrates is 1.0; for fats (tripalmitin) it is 0.7; for proteins it is 0.9; and for organic acids it is greater than 1.0.

• **Alert:** Remember that respiration is an amphibolic pathway because its intermediates are involved in both catabolism (breakdown) and anabolism (synthesis of biomolecules).

• **Alert:** Oxidation of one molecule of NADH produces 3 ATP molecules, while oxidation of one molecule of FADH2 produces 2 ATP molecules in the electron transport chain.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Respiratory Quotient (RQ). Write the formula and state the RQ values for carbohydrates and fats (tripalmitin). [Score: 2 Marks]

Answer & Key Valuation Steps:

• Definition: Respiratory Quotient is the ratio of the volume of CO2 evolved to the volume of O2 consumed in respiration. **[1 Mark]**

• Formula and values: RQ = Volume of CO2 evolved divided by Volume of O2 consumed; RQ for carbohydrates is 1.0, and for tripalmitin fat it is approximately 0.7. **[1 Mark]**

2. Why is the respiratory pathway considered an amphibolic pathway rather than purely catabolic? [Score: 2 Marks]

Answer & Key Valuation Steps:

• The respiratory pathway is catabolic because it breaks down respiratory substrates (carbohydrates, fats, proteins) to generate ATP energy. **[1 Mark]**

• It is also anabolic because respiratory intermediates are withdrawn for biosynthesis (e.g., Acetyl-CoA is used to synthesize fatty acids; oxaloacetate is used to synthesize amino acids); hence it is amphibolic. **[1 Mark]**

3. Trace the reactions of Glycolysis from glucose to the formation of fructose 1,6-bisphosphate and final pyruvic acid, highlighting ATP consumption and production steps. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Preparatory phase (ATP consumption): Glucose is phosphorylated to glucose 6-phosphate by hexokinase (consumes 1 ATP), isomerized to fructose 6-phosphate, and phosphorylated to fructose 1,6-bisphosphate by phosphofructokinase (consumes 1 ATP). **[1 Mark]**

• Cleavage and oxidation: Fructose 1,6-bisphosphate splits into two triose phosphates (PGAL and DHAP); PGAL is oxidized to 1,3-bisphosphoglycerate generating 2 NADH + 2H+. **[1 Mark]**

• Pay-off phase (ATP generation): 1,3-BPG transfers phosphate to ADP generating 2 ATP, and phosphoenolpyruvate (PEP) transfers phosphate to ADP generating 2 ATP by pyruvate kinase, yielding 2 pyruvic acid molecules and a net gain of 2 ATP. **[1 Mark]**

4. Describe the link reaction (oxidative decarboxylation of pyruvate) connecting glycolysis to the Krebs cycle, specifying the enzyme complex and products formed. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Location: Occurs in the mitochondrial matrix where pyruvate enters from the cytoplasm. **[0.75 Mark]**

• Enzyme complex: Catalyzed by the multi-enzyme pyruvate dehydrogenase complex requiring cofactors including NAD+ and Coenzyme A. **[0.75 Mark]**

• Reaction and products: Pyruvic acid + CoA + NAD+ yields Acetyl-CoA + CO2 + NADH + H+. **[0.75 Mark]**

• Significance: Generates Acetyl-CoA which serves as the primary entry metabolite for the Krebs citric acid cycle. **[0.75 Mark]**

5. Explain the arrangement and functions of the five enzyme complexes comprising the mitochondrial Electron Transport System (ETS). [Score: 3 Marks]

Answer & Key Valuation Steps:

• Complex I (NADH dehydrogenase): Oxidizes NADH from matrix and transfers electrons to ubiquinone, pumping protons into intermembrane space. **[0.6 Mark]**

• Complex II (Succinate dehydrogenase): Transfers electrons from FADH2 generated in Krebs cycle to ubiquinone without pumping protons. **[0.6 Mark]**

• Complex III (Cytochrome bc1 complex): Transfers electrons from ubiquinol to cytochrome c, pumping protons into intermembrane space. **[0.6 Mark]**

• Complex IV (Cytochrome c oxidase): Contains cytochromes a and a3 and two copper centers, transferring electrons to molecular oxygen to form water while pumping protons. **[0.6 Mark]**

• Complex V (ATP synthase): Couples proton flux from intermembrane space back into matrix through F0 channel to drive ATP synthesis at F1 headpiece. **[0.6 Mark]**

6. Differentiate between alcoholic fermentation and lactic acid fermentation with respect to enzymes, intermediate compounds, and release of carbon dioxide. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Enzymes: Alcoholic fermentation is catalyzed by pyruvate decarboxylase and alcohol dehydrogenase; lactic acid fermentation is catalyzed by lactate dehydrogenase. **[1 Mark]**

• Intermediate compounds: Alcoholic fermentation forms acetaldehyde intermediate; lactic acid fermentation directly converts pyruvate to lactic acid without an intermediate. **[1 Mark]**

• Carbon dioxide release: CO_2 is evolved during alcoholic fermentation; no CO_2 is released during lactic acid fermentation. **[1 Mark]**

7. Calculate the theoretical net ATP yield from the complete aerobic oxidation of one molecule of glucose through Glycolysis, Link reaction, Krebs cycle, and ETS. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Glycolysis: Direct substrate-level phosphorylation yields 2 ATP; 2 NADH oxidation via ETS yields 6 ATP (or 4 ATP depending on shuttle) = 8 ATP. **[1 Mark]**

• Link reaction: 2 molecules of pyruvate yield 2 NADH, which on oxidation in ETS yield 6 ATP. **[1 Mark]**

• Krebs cycle: 2 turns per glucose yield 2 GTP (equal to 2 ATP) via substrate phosphorylation; 6 NADH yield 18 ATP; 2 $FADH_2$ yield 4 ATP = 24 ATP. **[1 Mark]**

• Total net yield: 8 + 6 + 24 = 38 ATP molecules per molecule of glucose completely oxidized (theoretical balance sheet). **[1 Mark]**